Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience shipping **AI-powered full-stack products** across **Python**, **TypeScript**, **React**, and cloud services. Strong in fast end-to-end execution, LLM workflows, polished user-facing interfaces, and reliable backend systems for automation, integrations, and production release ownership.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University **Gandhinagar, In**
Bachelor of Technology in Computer Science and Engineering *November 2020 – May 2*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant **AI education platform** on **Azure SQL**, Blob Storage, and PostgreSQL for 500+ users, reducing dashboard latency from **1.2s to 350ms** through schema, query, and indexing optimization.
- Restructured a fault-tolerant **async LLM orchestration** system with **FastAPI** background workers, processing **1K+ AI jobs** with explicit job-state transitions and failure isolation while sustaining **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Engineered an **AI-powered Flask + React** grading platform using multimodal LLMs and OCR to evaluate **10K+ submissions**, with user-facing dashboards, rubric workflows, and production API integrations.
- Built concurrent backend processing and secure file-ingestion flows on **AWS RDS** with **GitHub Actions CI/CD**, handling **10K+ grading requests/day** and cutting turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with microservice architecture, automated VM testing, and **20+ unit/integration suites** using **Jenkins**, PyTest, and Mockito in an Agile Scrum environment.
- Optimized large-scale API integrations using **OOP patterns**, **AWS Glue**, and S3, improving response times by **30%** and pipeline efficiency by **68%** while reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **WebSockets**, REST APIs, and microservices, supporting **8+ business workflows** and synchronizing **2M+ records** with **33% fewer sync conflicts**.
- Automated deployments and release validation with **Jenkins**, Docker, and Azure SQL-backed services, reducing deployment time by **50%** while reaching **95%+ successful build validation**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **React Native**, **React**, **TypeScript**, Flask APIs, and Firebase sync across **50K+ records**, covering reservations, pricing, and webhook-driven payment flows.
- Mapped **12+ backend integrations** and shipped a PCI-compliant payment stack for cards, wallets, UPI, and PayPal, cutting manual payment handling by **70%** and improving end-to-end UX performance.

Skills

Languages: Python, JavaScript/TypeScript, Java, C++, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, JSON, Graph-based workflow logic
Frontend: React.js, Next.js, TailwindCSS, React Native, HTML, CSS
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Firebase
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions
AI/ML: LLM Systems, GPT-4o, Claude, OCR Workflows, Transformers, Prompt-Oriented Product Flows
Engineering Practices: CI/CD, Microservices, ETL Pipelines, OOP, Agile, System Design, Performance Optimization, Production Debugging

Projects

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, REST APIs*
- Built an **AI-assisted document platform** over **50K+ documents** with semantic search, summarization, and secure upload workflows, delivering sub-second retrieval and collaborative document interactions for 200+ users.

Chat2SQL | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace on top of Chat2DB, supporting **10+ databases** and accelerating query authoring, schema exploration, and structured execution from a unified **Java + TypeScript** product surface.

PiXMod | *Next.js, Transformers, MongoDB, Stripe*
- Built an AI SaaS platform for image workflows with transformer-based retrieval, generative editing, subscription billing, and responsive **Next.js** interfaces, processing **1K+ assets** while increasing premium conversion by **18%**.